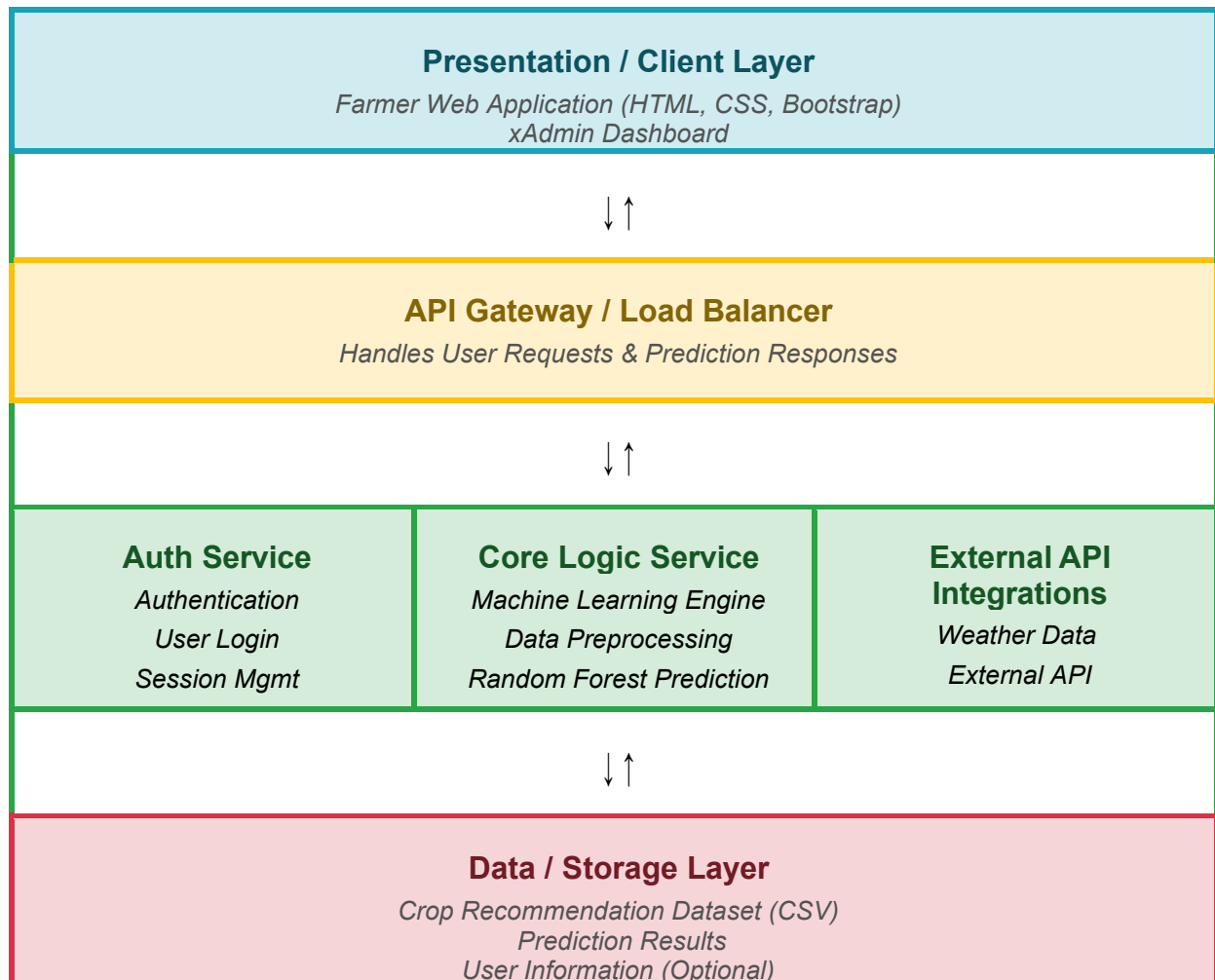


Solution Architecture

Date	28 JUNE 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Describe what this component does Provides a user-friendly web interface where farmers enter soil and environmental parameters and view crop recommendations.</i>	<i>HTML5, CSS3, Bootstrap, JavaScript</i>
<i>API Gateway/ Web Server</i>	<i>Receives user requests, validates inputs, communicates with the Machine Learning model, and returns prediction results.</i>	<i>Python, Flask</i>
<i>Core Logic Service (Machine Learning Engine)</i>	<i>Performs data preprocessing, feature selection, loads the trained model, and predicts the most suitable crop based on user input.</i>	<i>Scikit-learn, Pandas, NumPy</i>
<i>Authentication Service</i>	<i>Authenticates users and manages secure access to the application (if login functionality is implemented).</i>	<i>Flask Session / Python Authentication</i>
<i>External API Integration (Optional)</i>	<i>Retrieves weather or environmental information to improve prediction accuracy.</i>	<i>OpenWeather API or similar (Optional)</i>
<i>Data / Storage Layer</i>	<i>Stores the crop recommendation dataset, prediction results, and user-related information.</i>	<i>CSV Dataset, Pandas, Local Storage</i>